

Edge of Many-Body Quantum Chaos in Quantum Reservoir Computing

Kaito Kobayashi[✉] and Yukitoshi Motome[✉]

Department of Applied Physics, The University of Tokyo, Tokyo 113-8656, Japan



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Reservoir computing (RC) is a machine learning paradigm that harnesses dynamical systems as computational resources. In its quantum extension—quantum reservoir computing (QRC)—these principles are applied to quantum systems, whose rich dynamics broadens the landscape of information processing. In classical RC, optimal performance is typically achieved at the “edge of chaos,” the boundary between order and chaos. Here, we identify its quantum many-body counterpart using the QRC implemented on the celebrated Sachdev-Ye-Kitaev model. Our analysis reveals substantial performance enhancements near two distinct characteristic “edges”: a temporal boundary defined by the Thouless time, beyond which system dynamics is described by random matrix theory, and a parametric boundary governing the transition from integrable to chaotic regimes. These findings establish the “edge of many-body quantum chaos” as a design guideline for QRC.

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Introduction—Quantum mechanics harbors immense potential to revolutionize modern technology [1]. Breakthroughs in quantum computing, communication, and sensing have vividly showcased the transformative power of quantum resources [2–5]. These advances have, in turn, inspired extensive research efforts in quantum machine learning, with various approaches—notably variational quantum algorithms (VQAs)—being actively explored [6–11].

More recently, the power of quantum has also integrated with reservoir computing (RC), a machine learning framework renowned for its efficacy in time-series analysis [12–16]. True to its name, the most distinctive feature of RC is the use of an input-driven dynamical system, known as a “reservoir,” to nonlinearly map input data onto a feature space. Notably, training in RC is limited to the postprocessing stage, and the reservoir itself remains fixed, which significantly reduces the training cost. Extending this idea into the quantum domain, Fujii and Nakajima introduced quantum reservoir computing (QRC) [17]. QRC harnesses natural quantum dynamics as a reservoir, generating rich, nonlinear input-to-output mappings in an exponentially large Hilbert space without encountering the trainability bottlenecks prevalent in VQAs. This capability has sparked significant theoretical progress [18–38], as well as promising experimental demonstrations [39–47].

Given the optimization-free nature of RC, researchers naturally ask what characterizes an effective reservoir system. While this question remains largely unanswered for QRC, several key criteria have been identified for classical RC. The most renowned among them is the concept of the “edge of chaos” [48–52]. This posits that the boundary between chaotic and nonchaotic dynamical regimes provides a performance sweet spot for RC systems,

as empirically verified across diverse platforms [53–55]. By analogy, exploring a quantum version of the edge of chaos offers a promising route to addressing the fundamental question posed above.

In classical systems, chaos is naturally framed via phase space trajectories [56]. In quantum mechanics, however, the Heisenberg uncertainty principle precludes such a direct picture, and one must instead rely on more subtle diagnostics. Early investigations of quantum chaos therefore focused on single-particle systems whose classical limits are chaotic, leading to the conjecture (with some analytical support) that their spectral statistics are governed by random matrix theory (RMT) [57–61]. In recent years, attention has shifted to many-body settings, spurred in large part by the exactly solvable Sachdev-Ye-Kitaev (SYK) model [62–65]. Various diagnostics, including level statistics [66–69], out-of-time-order correlators [70–72], and spectral form factors [73–76] have enriched our understanding of chaos in interacting quantum systems. Experimental realizations of the SYK physics have also been proposed on several platforms [77–81]. Notably, the SYK model exhibits maximally quantum chaotic behavior, saturating the so-called bound on chaos [72,82,83] and has stood as a canonical model for the investigation of many-body quantum chaos. These insights motivate the exploration of QRC built upon the SYK model, aiming to address the open question of reservoir effectiveness from a quantum chaotic standpoint.

In this Letter, we present a unified, RMT-based analysis that identifies the “edge of many-body quantum chaos” as the optimal guide for operating QRC. In particular, we examine the QRC performance near two distinct chaotic boundaries of the SYK model. The first is a temporal boundary defined by the Thouless time, beyond which the

system's dynamics conforms to RMT. The second is a parametric boundary marking the transition from quantum chaotic to integrable (nonchaotic) behavior, controlled by the strength of an additional noninteracting term. Using representative machine learning tasks that demand both memory retention and nonlinear transformation, we demonstrate that QRC performance is maximized near the onset of quantum chaotic regime in both temporal and parametric domains. These observations establish the edge of many-body quantum chaos as an important guideline for the design of QRC systems.

Model—Figure 1 schematically illustrates the QRC architecture implemented with the SYK model for fermions. The quantum reservoir system, whose dynamics is harnessed for information processing, is governed by the Hamiltonian

$$\mathcal{H} = \sum_{i,j,k,l=1}^N J_{ijkl} c_i^\dagger c_j^\dagger c_k c_l + \sum_{i,j=1}^N \kappa_{ij} c_i^\dagger c_j, \quad (1)$$

where N denotes the system size, and $c_i^\dagger$ (c_i) is the spinless fermionic creation (annihilation) operator at site i . In

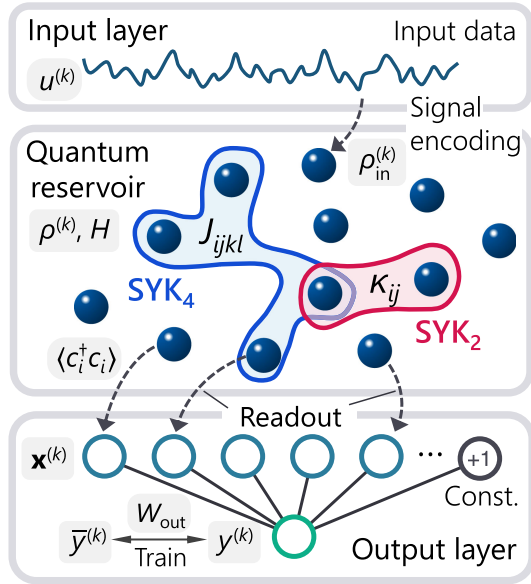


FIG. 1. Schematic illustration of the SYK model and its implementation in QRC architecture. The SYK system described in Eq. (1), which serves as the quantum reservoir, comprises the SYK₄ term (J_{ijkl}) and the SYK₂ term (κ_{ij}). At each time step k , the process begins by encoding an input $u^{(k)}$ into the quantum reservoir via the input state $\rho_{\text{in}}^{(k)}$. The reservoir state $\rho^{(k)}$ then evolves under the Hamiltonian according to the update rule in Eq. (2). Information is subsequently read out by constructing a state vector $\mathbf{x}^{(k)}$ from the expectation values $\langle c_i^\dagger c_i \rangle$ and a constant bias term. Finally, this vector is linearly transformed by the trainable weight matrix W_{out} to produce the output $y^{(k)}$, which is optimized to approximate the target $\bar{y}^{(k)}$.

Eq. (1), the first term describes the SYK₄ interaction: the all-to-all couplings J_{ijkl} are drawn from a zero-mean complex Gaussian distribution of variance $\overline{|J_{ijkl}|^2} = J_4^2/N^3$ ($\bar{\bullet}$ denotes the ensemble average). The second term is the noninteracting SYK₂ contribution, with coupling constants κ_{ij} sampled from a zero-mean complex Gaussian distribution of variance $\overline{|\kappa_{ij}|^2} = \kappa_2^2/(2N)$. The couplings satisfy the symmetry relations $\kappa_{ij} = \kappa_{ji}^*$, $J_{ijkl} = J_{klij}^*$, and $J_{ijkl} = -J_{jikl} = -J_{ijlk}$. Both coupling terms conserve the total fermion number $N_p \equiv \sum_i \langle c_i^\dagger c_i \rangle$. Unless otherwise specified, we set $N = 8$, $J_4 = 1$, and $N_p = N/2$ (note that the QRC input protocol introduced below does not conserve N_p).

To diagnose the chaotic nature of quantum many-body systems, we first examine the level statistics [66–69]. Let the ordered eigenvalues of $\mathcal{H}$ be E_n ; the spacings between consecutive energy levels are given by $s_n = E_{n+1} - E_n$. According to RMT, the spacing distribution in integrable systems follows Poisson statistics, whereas in quantum chaotic systems, it is well described by Wigner-Dyson (WD) distribution. A convenient metric to characterize these distributions is the level spacing ratio, defined as $r_n = \min(s_n/s_{n+1}, s_{n+1}/s_n)$. Its mean value $\langle r \rangle$ assumes ≈ 0.3862 for Poisson distribution and ≈ 0.5996 for WD statistics [66,67]. Figure 2(a) presents the distribution of the level spacing ratio $P(r)$ for the SYK₄ model [$(J_4, \kappa_2) = (1, 0)$] and the SYK₂ model [$(J_4, \kappa_2) = (0, 1)$], in excellent agreement with the WD and Poisson predictions, respectively. Notably, when both terms are present, the system can be tuned between these extremes: as the ratio of the typical coupling strengths κ_2/J_4 increases, Fig. 2(b) displays a gradual shift of $\langle r \rangle$ from the WD to the Poisson value. This result strongly suggests a transition from quantum-chaotic to integrable behavior [84–88]; for finite-size effects, see Supplemental Material [89].

Framework of the QRC—We now describe the QRC framework for information processing; all procedures employed herein follow the conventional protocols established in earlier investigations [17,18]. The QRC architecture comprises four principal stages: signal encoding, reservoir dynamics, readout, and postprocessing, through which the input at the k th step $u^{(k)}$ is processed in an attempt to produce its corresponding target value $\bar{y}^{(k)}$ (Fig. 1). Consider a discretized input sequence $\{u^{(k)}\} \equiv \{u^{(0)}, u^{(1)}, \dots\}$ with $u^{(k)} \in [0, 1]$. Each input $u^{(k)}$ is encoded into the reservoir by replacing the quantum state at site 1 with $\rho_{\text{in}}^{(k)} \equiv |\psi_{\text{in}}^{(k)}\rangle\langle\psi_{\text{in}}^{(k)}|$, where $|\psi_{\text{in}}^{(k)}\rangle \equiv \sqrt{1-u^{(k)}}|0\rangle + \sqrt{u^{(k)}}|1\rangle$. Subsequently, the system evolves unitarily under the Hamiltonian in Eq. (1). The encoding operation is repeated at intervals Δt_{in} , which determines the intrinsic timescale of the QRC system. The overall update rule is summarized by the completely positive trace-preserving map:

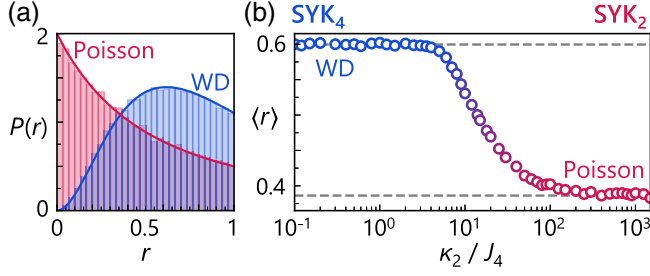


FIG. 2. (a) Distribution of level spacing ratio $P(r)$ for the SYK₄ model (blue) and the SYK₂ model (red). Corresponding WD and Poisson predictions are shown in blue and red curves, respectively. (b) Level spacing ratio $\langle r \rangle$ as a function of the coupling strength κ_2/J_4 , computed over the central 50% of the spectrum. The horizontal dashed lines represent the reference values for Poisson and WD statistics. In both panels, data are averaged over 2000 realizations for the system size $N = 8$.

$$\rho^{(k)}(t) = e^{-i\mathcal{H}t} [\rho_{\text{in}}^{(k)} \otimes \text{Tr}_1 \{ \rho^{(k-1)}(\Delta t_{\text{in}}) \}] e^{i\mathcal{H}t}, \quad (2)$$

where $\text{Tr}_1 \{ \bullet \}$ denotes the partial trace over site 1; see Supplemental Material for details on the observed dynamics [89]. During the evolution, the expectation value of the single-site particle number is measured at discrete times $t_v = v\Delta t_{\text{in}}/V$, where $v = 1, \dots, V$ and V is the number of virtual computational nodes (here, $V = 10$); for the effect of statistical noise, see Supplemental Material [89]. At the k th step, the outcomes $\text{Tr} \{ \rho^{(k)}(t_v) c_i^\dagger c_i \}$ are assembled into a one-dimensional state vector $\mathbf{x}^{(k)}$; including an additional constant term, $\mathbf{x}^{(k)}$ has $NV + 1$ components in total. The output layer then applies a linear transformation $y^{(k)} = W_{\text{out}} \mathbf{x}^{(k)}$, where the output weight W_{out} is trained by minimizing the mean squared error between the computed output $y^{(k)}$ and the target output $\bar{y}^{(k)}$. In our simulations, a total of 10 000 inputs are processed. Starting from a random mixed state, the first 4000 inputs are discarded to wash out the effects of initial conditions (see Supplemental Material for the convergence property [89]). The subsequent 3000 inputs are employed for training W_{out} , and the final 3000 are reserved for evaluation. Performance results are averaged over 500 independent random realizations.

Benchmark tasks in RC are typically characterized by their requirements on temporal memory and nonlinear processing: the former specifies how far back in the input history the reservoir must retain information, while the latter is often described by the maximum polynomial order and the richness of cross-term interactions. Accordingly, we evaluate the reservoir performance on two complementary tasks: the short-term memory (STM) task, which probes linear memory capacity, and the nonlinear autoregressive moving average (NARMA) task, which evaluates combined memory and nonlinear processing capabilities [14,17]. Since real-world applications hinge on both properties, we place greater emphasis on the NARMA

performance; for additional tasks, see Supplemental Material [89].

In the STM task, the target output at delay d is defined as the input value d steps earlier in a random input sequence $\{u^{(k)}\}$, i.e., $\bar{y}_d^{(k)} = u^{(k-d)}$. During the testing phase, we collect the outputs $\mathbf{y} = \{y^{(k)}\}$ and targets $\bar{\mathbf{y}}_d = \{\bar{y}_d^{(k)}\}$, and compute the determination coefficient $R_d^2 \equiv \text{cov}^2(\mathbf{y}, \bar{\mathbf{y}}_d) / [\sigma^2(\mathbf{y})\sigma^2(\bar{\mathbf{y}}_d)]$, where cov and σ^2 denote the covariance and the variance, respectively. R_d^2 is bounded between 0 and 1; values closer to 1 indicate better memory performance.

In the NARMA task, the target sequence at a given order n is defined by the recursive relation: $\bar{y}^{(k+1)} = \alpha \bar{y}^{(k)} + \beta \bar{y}^{(k)} \sum_{j=0}^{n-1} \bar{y}^{(k-j)} + \gamma u^{(k-n+1)} u^{(k)} + \delta$, where $(\alpha, \beta, \gamma, \delta)$ are set to (0.3, 0.05, 1.5, 0.1) [17,20]. The input stream is sampled from the range $[0, 0.2]$ to prevent divergence in $\{\bar{y}^{(k)}\}$, which is then rescaled to $[0, 1]$ when encoded to the quantum reservoir [17]. Performance is evaluated in terms of the deviation between the output and the target, which is quantified by the normalized mean-squared error (NMSE) $\|\mathbf{y} - \bar{\mathbf{y}}\|^2 / \|\bar{\mathbf{y}}\|^2$ with $\|\bullet\|$ being the Euclidean norm. A lower NMSE indicates superior performance in accurately reproducing the target nonlinear dynamics.

Temporal edge of many-body quantum chaos—We begin with an examination of the QRC performance of the SYK₄ and SYK₂ models, in which clear fingerprints of many-body quantum chaos manifest themselves in the temporal domain. As a time-domain probe of the energy-spectrum correlations characteristic of quantum chaos, we analyze the spectral form factor (SFF) $K(t) = \overline{\sum_{m,n} e^{i(E_m - E_n)t}} / \mathcal{N}^2$, where $\mathcal{N}$ denotes the Hilbert space dimension [73–76]. In many-body chaotic systems, the SFF generally exhibits a “ramp-plateau” profile governed by two timescales: the Thouless time t_{Th} , after which the evolution follows RMT predictions, leading to a universal “ramp” reflecting level repulsion, and the plateau time t_p , beyond which $K(t)$ saturates at a plateau due to the discreteness of the spectrum [90–92]. Figures 3(a) and 3(b) illustrate the SFF for the SYK₄ and SYK₂ limits, respectively. In the SYK₄ case, the SFF exhibits the pronounced ramp beginning around $t_{\text{Th}} \sim 4.5$ and the subsequent plateau beyond $t_p \sim 80$, providing a clear signature of many-body quantum chaos. While $K(t)$ in the SYK₂ model also features a dip and a plateau, they occur on a much shorter timescale and with a different system-size scaling of t_p compared to the chaotic system (see Supplemental Material for details [89]).

Figures 3(c) and 3(d) plot the QRC performance of the SYK₄ and SYK₂ models as functions of the input interval Δt_{in} , which defines the intrinsic timescale governing the QRC dynamics. The upper panels display the STM performance $R_{d=n-1}^2$ at delay $d = n - 1$, and the lower panels show the NMSE on the NARMA tasks of

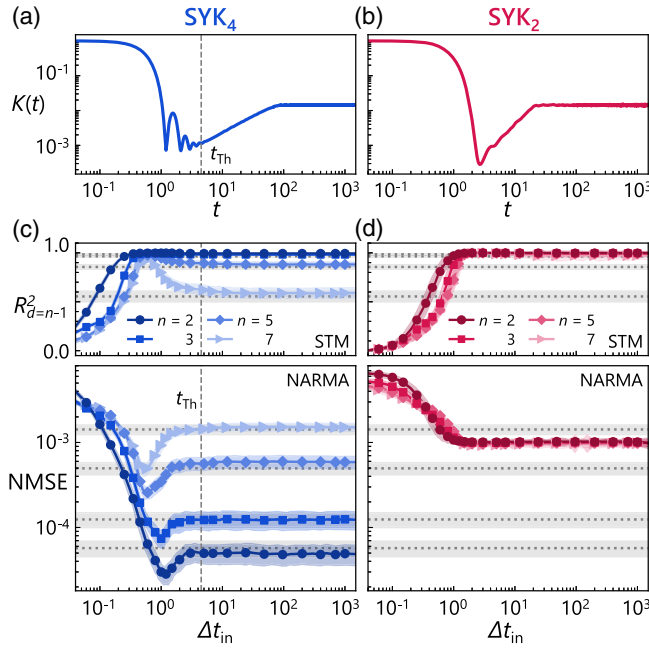


FIG. 3. (a),(b) The SFF $K(t)$ for the SYK_4 and SYK_2 models, respectively, with each curve averaged over 20 000 realizations. (c),(d) The QRC performances as functions of Δt_{in} for the SYK_4 and SYK_2 models. The upper panels show the memory performance $R^2_{d=n-1}$ (higher values indicate better retention), while the lower panels display the NMSE on the NARMA tasks (lower values indicate better accuracy). Markers represent the orders $n = 2, 3, 5$, and 7 (plotted sparsely for brevity). The horizontal dotted lines indicate the reference performance of the Haar QRC model, with poorer performance corresponding to higher n . Performance results are averaged over 500 realizations, with shaded regions denoting the standard deviation. Vertical dashed lines in (a) and (c) represent the Thouless time t_{Th} .

orders $n = 2, 3, 5$, and 7 . The chosen STM delay corresponds to the oldest input term explicitly appearing in the definition of the NARMA- n dynamics; additional past inputs are nevertheless recursively encoded within the target sequence $\{\bar{y}^k\}$ (see Supplemental Material for the full delay dependence of R^2_d [89]). To establish a baseline for comparison, we also analyze the Haar QRC model, in which the evolution operator $e^{-i\mathcal{H}t}$ is replaced by a unitary U drawn from the Haar measure; no intrinsic timescale exists therein [93,94]. Its performance, shown as horizontal dashed lines, deteriorates monotonically with increasing n as the task complexity grows.

A striking pattern emerges in the QRC performance for the SYK_4 model [Fig. 3(c)]. When Δt_{in} is very small, the time evolution operator of the quantum reservoir is nearly the identity, failing to enrich the input-to-output map and resulting in poor performance on both the STM and NARMA tasks. However, as Δt_{in} increases toward the Thouless time t_{Th} , the QRC performance improves dramatically and reaches its maximum just before t_{Th} . Beyond this point, both metrics level off at the baseline of the Haar

QRC model for each order n , in accordance with the onset of universal RMT descriptions typical of quantum chaotic systems. In other words, the Thouless time effectively marks the temporal boundary where fully quantum chaotic signatures begin to dominate the QRC performance. The observed dip of the NMSE, situated just prior to t_{Th} , thus explicitly underscores the performance enhancement at the edge of many-body quantum chaos in the temporal domain.

In contrast, the QRC with the integrable SYK_2 model behaves quite differently [Fig. 3(d)]. As Δt_{in} increases, the NARMA performance improves only modestly from its initial low values and quickly saturates, showing no pronounced peak. This behavior starkly contrasts with that of the SYK_4 model, indicating that the performance enhancements observed in Fig. 3(c) are fundamentally attributable to the latter's intrinsic many-body quantum chaotic nature. Notably, despite its poor NARMA performance, the SYK_2 model attains higher STM performance than the SYK_4 model. These model-dependent behaviors motivate our exploration of interpolations between the SYK_2 and SYK_4 models.

Parametric edge of many-body quantum chaos—We here demonstrate the QRC performance across the quantum chaotic-integrable transition by mixing the SYK_4 and SYK_2 Hamiltonians. To ensure a fair comparison as the coupling strength is varied, each Hamiltonian is rescaled by its spectral norm: $\mathcal{H} \rightarrow \mathcal{H}/\|\mathcal{H}\|$ [95]. (Note that this correspondingly changes the timescale relative to Fig. 3.) Figures 4(a) and 4(b) display the STM and NARMA performances versus the ratio κ_2/J_4 for long ($\Delta t_{\text{in}} = 50$) and short ($\Delta t_{\text{in}} = 1$) input intervals, respectively. The vertical dashed lines serve as a guide of the eye for the onset of WD and Poisson statistics, while the horizontal dotted lines represent the performance of the Haar QRC model for reference. We note that these input intervals are chosen to probe two distinct timescales: the longer input interval $\Delta t_{\text{in}} = 50$ evaluates QRC performance within the universal RMT regime, while the shorter interval $\Delta t_{\text{in}} = 1$ characterizes QRC behavior in the nonuniversal, short-time window outside the scope of RMT.

The most pronounced trend is observed for $\Delta t_{\text{in}} = 50$ [Fig. 4(a)]. When κ_2/J_4 is varied, the NARMA performance exhibits a clear dip in the NMSE at a coupling strength just before the onset of the many-body quantum chaotic regime. This dip provides another manifestation of the edge of many-body quantum chaos, in this case in the parameter domain. By contrast, for the short input interval $\Delta t_{\text{in}} = 1$, varying the coupling strength yields no performance enhancement [Fig. 4(b)]. At such a short timescale, the QRC does not reflect chaotic characteristics, so the performance changes only gradually, without any distinct peak at the chaos boundary.

Discussions—In the strongly chaotic regime with large Δt_{in} , the QRC provides a highly nonlinear mapping, but the

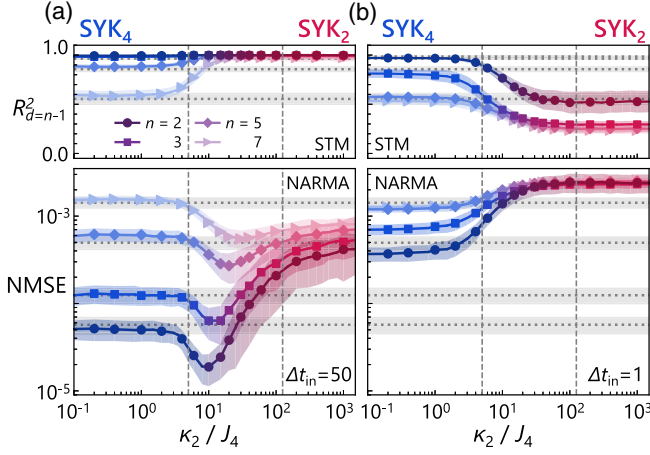


FIG. 4. (a),(b) The QRC performance as a function of κ_2/J_4 for the input interval $\Delta t_{\text{in}} = 50$ and $\Delta t_{\text{in}} = 1$, respectively, analogous to Figs. 3(c) and 3(d). Sparsely placed markers denote the order n , with colors indicating $\langle r \rangle$ according to the color scale in Fig. 2(b). The vertical dashed lines indicate the approximate boundaries where the absolute difference between the measured $\langle r \rangle$ and its theoretical value for WD (left) and Poisson (right) statistics first falls below 10^{-2} .

intrinsic featurelessness of the quantum chaotic dynamics makes input-specific signatures relatively indistinct. Conversely, in the nonchaotic regime or with moderately small Δt_{in} , the QRC retains input memory but sacrifices nonlinear expressivity. At the edge of many-body quantum chaos, the expressivity of the input-to-output map is optimally balanced against the preservation of input-specific features, leading to enhanced performance. From this picture, it follows that the precise location of this edge and the achievable gain depend on the task. Tasks that demand long memory shift the performance peak toward the nonchaotic side (e.g., the STM task with large d), whereas tasks that require stronger nonlinearity at fixed memory shift it toward the chaotic side (e.g., the nonlinear STM task). In extreme cases—when a task overwhelmingly requires either memory or nonlinearity, or when its complexity exceeds the QRC’s capacity—the edge-induced performance enhancement may not appear. See Supplemental Material for additional performance evaluations [89].

Recent advances in QRC have reported related performance enhancements, and we believe that a concise comparison highlights the significance of our contributions. One of the authors and his collaborators identified a performance peak near the operational regime boundary in their feedback-driven QRC protocol [32]; however, this enhancement appears predominantly due to classical instabilities in feedback connections, rather than intrinsically quantum phenomena. In the purely quantum regime, Martínez-Peña *et al.* demonstrated enhanced performance near the ergodic-localization transition in the spin-based QRC [20]; see also Refs. [23,33]. While this finding

highlights the importance of dynamical phases in QRC, the role of many-body quantum chaos, and its relation to RMT, remain largely unexplored. Recently, Llodrà *et al.* analyzed QRC performance in connection with quantum chaos appearing at the quantum phase transition in a clean system, which is complementary to our Letter on random systems [34]. From a standpoint of temporal domain, investigations into quantum chaos remain sparse: Vetrano *et al.* explored the role of scrambling times, though within a different framework (quantum extreme learning machines) [94].

Building upon these studies, our Letter directly demonstrates the relationship between quantum chaos and QRC. Using the SYK model—a canonical many-body quantum chaotic system—as a test bed, we have extended the concept of the edge of chaos into the quantum many-body realm by uncovering two distinct edges: one is the temporal edge, which emerges before the Thouless time, and the other is the parametric edge, which appears near the quantum chaotic-to-integrable transition. Since our RMT-based analysis makes no system-specific assumptions, we expect the observed edge of many-body quantum chaos to be broadly observable across platforms. Our Letter therefore provides clear design guidelines for engineering quantum reservoirs, unlocking advanced information processing with quantum many-body systems.

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Data availability—The data that support the findings of this article are not publicly available upon publication because it is not technically feasible and/or the cost of preparing, depositing, and hosting the data would be prohibitive within the terms of this research project. The data are available from the authors upon reasonable request.

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